

IMPERIAL

Department of Computing

70015: Mathematics for Machine Learning

Lecturer: Matthew Wicker

Lecture 00: Preliminaries and Background

Additional Resources

- Linear Algebra: Chapter 2 (page 29) [Bengio et al. \(2017\)](#)
https://www.deeplearningbook.org/contents/linear_algebra.html
- Probability Theory Review: Chapter 2 (page 27) [Murphy \(2012\)](#)
- Calculus Review: Chapter 5 (page 139) [Deisenroth et al. \(2020\)](#)

0.1 Course Introduction

This course aims to introduce students to a formal mathematical perspective on machine learning. The course assumes students have a basic level of prerequisite knowledge in linear algebra, calculus, and statistics. This lecture note, [Lecture 00](#), goes over some basic fundamentals and provides links to more comprehensive resources.

There are Practice Problems 00 and Solutions 00 which students are welcomed to use to brush up on background material. These questions are not curated to provide targeted background for the material we will cover, but they do provide good coverage of fundamentals.

0.1.1 Course Materials & Assessments

Before each lecture, notes, slides, and exercise questions will be posted on Scientia. Model solutions will be posted (roughly) the day after the lecture in order to provide students with ample time to solve the questions on their own. These questions do not count for a grade and are primarily for students to practice the techniques we observe during lecture and are highly encouraged.

Lecture notes will *not* be provided more than a day in advance of our lecture. This is intentional. Every iteration of the course is adapted in pace and material. This is done to keep the material as up to date as possible and to allow us to re-explain any materials by popular demand. If you would like to prepare in advance for the material we will cover, I do provide last year's complete notes under the filename PriorMaterials.pdf. **NB: Course material changes about 10% in any given year. You are responsible for the material we cover *this* year.**

The two activities that are marked are the course project and final exam. The course project mirrors an academic workshop and challenges students with a synthetic dataset on which they must apply the course content and write up a report. This will count for 30% of the overall course grade. The second assessment is the final exam. Examples of the final exam are hosted by Imperial. Additionally, this year the end of each lecture will contain a practice exam question that we will go through together. Further, the final lecture of class will be dedicated to solving practice exams to prepare students. Some material discussed during the course will not appear on the exam. These sections will be marked with a red star (★) at the top of the section.

0.1.2 Course Syllabus

The below list of content is a tentative course syllabus that is subject to change with any unforeseen circumstances.

- [Lecture 00★](#) - Exercises in linear algebra, calculus, and statistics
- [Lecture 01](#) - ML Problem Settings & Linear Models
- [Lecture 02](#) - Ordinary Least Squares & Generalized Linear Models
- [Lecture 03](#) - Gradient Descent & Hessians
- [Lecture 04](#) - Deep Learning & Automatic Differentiation
- [Lecture 05](#) - Convexity & Convergence
- [Lecture 06](#) - Concentration Inequalities & Generalization Bounds
- [Lecture 07](#) - Statistical Estimators & Bias-Variance Trade-off
- [Lecture 08](#) - Maximum Likelihood for Unsupervised Learning
- [Lecture 09](#) - Maximum Likelihood for Supervised Learning
- [Lecture 10](#) - Bayesian Machine Learning, Part I
- [Lecture 11](#) - Bayesian Machine Learning, Part II
- [Lecture 12](#) - Representation Learning & Dimensionality Reduction
- [Lecture 13★](#) - Learning and Language Modeling
- [Lecture 14★](#) - Trustworthiness in Machine Learning & Exam Preparation

0.1.3 Course Logistics (2026)

Course lectures will be held in Huxley room 308 and will be recorded on Panopto for later viewing, though students are encouraged to attend in person where possible. Lectures are held on Tuesday from 9 a.m. - 11 a.m. and Thursday from 9am - 11 a.m. every week of term unless indicated otherwise. Though the lectures are only one hour, the subsequent hour is reserved for problem sessions. This year's problem sessions will follow a new pilot structure:

1. Initial break, individual study, and questions for Matthew to work on the whiteboard.
2. Practice exam question completed individually or quietly in a group.
3. Model answer walkthrough and questions.

Matthew Wicker will hold office hours for this class between 2 p.m. and 4 p.m. on Thursdays. This time is reserved solely for students of this class and is intended both for content clarification and any research inquiries. Please come to ACEX 258 during the given time window if you would like to attend.

0.1.4 Course Materials

The primary course materials include (1) lecture notes, (2) lecture slides, (3) recordings, (4) exercises and solutions, (5) practice exam questions and solutions, and (6) the referenced textbooks [Deisenroth et al. \(2020\)](#); [Murphy \(2012\)](#); [Bishop \(2006\)](#).

While these lecture notes will serve as the primary guide for this course, it is important to supplement any doubts or questions with more comprehensive treatments by reference textbooks. Each lecture note provides pointers to sections from the following three books that are publically available at the specified urls.

- The textbook, *Mathematics for Machine Learning*, to be referenced as [Deisenroth et al. \(2020\)](#), can be found at: <https://mml-book.github.io/book/mml-book.pdf>
- The textbook, *Machine Learning a Probabilistic Perspective*, to be referenced as [Murphy \(2012\)](#), can be found at: <https://raw.githubusercontent.com/kerasking/book-1/master/ML%20Machine%20Learning-A%20Probabilistic%20Perspective.pdf>
- The textbook, *Pattern Recognition and Machine Learning*, to be referenced as [Bishop \(2006\)](#), can be found at: <https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf>

0.2 Course Notation

This course must assume some mathematical maturity and familiarity. Some of the below terms may be unfamiliar to you. Some of the treatments may seem sparse. If this is the case for you, please use referenced materials, graduate teaching assistants, or your peer group.

Throughout this course we will denote real values with lower-case Roman letters e.g., x . Vectors will be denoted with lower-case bold Roman or Greek letters, e.g., $\mathbf{x}$. This should not be confused with random variables which are denoted with bold, non-italicized Roman letters: $\mathbf{x}$. Matrices will be denoted with upper-case Roman letters, e.g., $\mathbf{X}$, and random matrices are denoted $\mathbf{X}$. Greek letters will always have their type specified with the exception of θ being used, without loss of generality, as the collection of all model parameters into a single vector.

Course conventions with respect to vector operations and vector calculus are given in their respective sections below, and should be reviewed by all students.

0.3 Linear Algebra

A core prerequisite for this class is linear algebra. Please use this section and references herein to brush up on necessary concepts.

The two key objects in all of linear algebra are vectors and matrices. A vector, denoted $\mathbf{x} \in \mathbb{R}^n$, is described as being n -dimensional. From a programmer's point of view, we can imagine this as a list of n real numbers, and we can access each dimension of the vector via a subscript i.e., $\mathbf{x}_i$ referring to the value of the i^{th} dimension of the vector. Matrices, denoted $\mathbf{X} \in \mathbb{R}^{n \times m}$ can be viewed as 2D arrays with dimensions $n \times m$ which we say has n rows and m columns.

It is often that we will want to multiply matrices and vectors together. In such cases, it is convient to view vectors as matrices with either $n = 1$ which we call a row vector (given that it represents 1 row), or $m = 1$ which we call a column vector (given that it represents 1 column). Unless otherwise stated, students should always assume that vectors are column vectors, i.e., take the shape $n \times 1$. An example showing how a vector is indexed is illustrated below:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$$

We can visualize the matrix indeces as follows:

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_{1,1}, \mathbf{X}_{1,2}, \dots, \mathbf{X}_{1,m}, \\ \mathbf{X}_{2,1}, \mathbf{X}_{2,2}, \dots, \mathbf{X}_{2,m}, \\ \vdots, \ddots, \vdots \\ \mathbf{X}_{n,1}, \mathbf{X}_{n,2}, \dots, \mathbf{X}_{n,m} \end{bmatrix}$$

Note that each entry of a vector, $\mathbf{x}_i$, is a scalar and so may appear equivalently as x_i in your solutions. To multiply two matrices together, the columns of the left hand term must equal the rows of the right hand term. For example if we would like to multiply a matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$ with a matrix $\mathbf{B} \in \mathbb{R}^{m \times k}$ this is allowed. However, if we cannot multiply $\mathbf{A}$ by a matrix $\mathbf{C} \in \mathbb{R}^{k \times m}$. One way to ensure that you are able to multiply two matrices is to recall the definition of matrix multiplication. The result of multiplying $\mathbf{A}$ by $\mathbf{B}$ is a matrix $\mathbf{C}$ with elements defined by:

$$C_{i,j} = \sum_{k=1}^m A_{i,k} B_{k,j}$$

Notice that the index k in the summation ranges from 1 to m meaning that $\mathbf{A}$ must have exactly m columns and $\mathbf{B}$ must have exactly m rows.

We complete our linear algebra review by recalling standard notations we will use throughout the course. Here is some standard notation for vectors. (We assume $\mathbf{u}$ and $\mathbf{v}$ are both n -dimensional vectors.)

Symbol	Meaning
$\mathbf{v}^\top$	Transpose of $\mathbf{v}$
$\mathbf{u}^\top \mathbf{v}$	Inner (scalar) product
$\mathbf{u}\mathbf{v}^\top$	Outer product ($n \times n$ matrix)
$\mathbf{u} \odot \mathbf{v}$	Elementwise product, $[\mathbf{u}_1 \mathbf{v}_1, \dots, \mathbf{u}_n \mathbf{v}_n]$
$\dim(\mathbf{v})$	Dimensionality of $\mathbf{v}$ (namely n)
$\text{diag}(\mathbf{v})$	Diagonal $n \times n$ matrix made from vector $\mathbf{v}$
$\mathbf{1}$ or $\mathbf{1}_n$	Vector of ones (of length n)
$\mathbf{0}$ or $\mathbf{0}_n$	Vector of zeros (of length n)
$\ \mathbf{v}\ $ or $\ \mathbf{v}\ _2$	Euclidean or ℓ_2 norm, $\sqrt{\sum_{i=1}^n \mathbf{v}_i^2}$
$\ \mathbf{v}\ _1$	ℓ_1 norm, $\sum_{i=1}^n \mathbf{v}_i $
$\mathbf{e}_i$	i^{th} -basis vector (1 in dimension i and 0 else where).

Standard notation for matrices, (we assume $\mathbf{S}$ is a square $n \times n$ matrix, $\mathbf{X}$ and $\mathbf{Y}$ are of size $m \times n$, and $\mathbf{Z}$ is of size $m' \times n'$.)

Symbol	Meaning
$\mathbf{X}_{:,j}$	j -th column of matrix
$\mathbf{X}_{i,:}$	i -th row of matrix (treated as a column vector)
$\mathbf{X}_{ij}$	Element (i, j) of matrix
$\mathbf{X}^\top$	Transpose of a matrix
$\text{diag}(\mathbf{S})$	Diagonal vector extracted from square matrix
$\mathbf{I}$ or $\mathbf{I}_n$	Identity matrix of size $n \times n$
$\mathbf{X} \odot \mathbf{Y}$	Elementwise product
$\mathbf{X} \otimes \mathbf{Z}$	Kronecker product
$\mathbf{S} \succ 0$	True iff $\mathbf{S}$ is a positive definite matrix
$\text{tr}(\mathbf{S})$	Trace of a square matrix
$\det(\mathbf{S})$	Determinant of a square matrix
$ \mathbf{S} $	Determinant of a square matrix

0.4 Calculus

In this section, we cover some basic multi-variate calculus and useful identities. We will review a few of the most pertinent of these identities in future lectures; however, this is considered prerequisite material and so students should spent time familiarizing themselves with this content if they find it challenging or new. For those who simply need a more in-depth refresher please refer to Chapter 5 of [Deisenroth et al. \(2020\)](#).

Unqualified, we refer to just a *function* as a mapping from the reals to the reals: $f : \mathbb{R} \rightarrow \mathbb{R}$. As usual, the mapping of a value x from the domain is denoted $f(x)$. We then define the derivative as:

$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (1)$$

assuming this limit exists. The above form is also known as the forward finite difference quotient. We will often use *Leibniz* notation to denote the derivative of a function $y = f(x)$ as $\frac{dy}{dx}$ or $\frac{d}{dx}f$ and use the notation $\frac{dy}{dx}|_a$ to denote the derivative of the function f at a . It will sometimes prove useful to use *Lagrange* notation where the derivative is simply f' and the derivative at a is simply $f'(a)$.

0.4.1 Multivariate Calculus

In this course, we will often care about the derivatives of functions with vector arguments i.e., $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We call such functions *vector-valued* functions and we define the partial derivative to be:

$$\frac{df}{dx_i} = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}, \quad (2)$$

again assuming the limit exists. Building on this definition, we define the gradient of a function to be the collection of all of its partial derivatives into a single *row* vector:

$$\mathbf{g} = \nabla f = \frac{\partial f}{\partial \mathbf{x}} = \left[\frac{df}{dx_0}, \frac{df}{dx_1}, \dots, \frac{df}{dx_n} \right] \quad (3)$$

Notice here that $\mathbf{g}$ is a function that maps vectors to vectors: $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ which we call a *vector field*. The derivative of an arbitrary vector field $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a matrix that we call the *Jacobian*:

$$\nabla f = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial \mathbf{x}} \\ \frac{\partial f_2}{\partial \mathbf{x}} \\ \vdots \\ \frac{\partial f_m}{\partial \mathbf{x}} \end{bmatrix} \quad (4)$$

where we use f_i to denote the i^{th} output dimension of the function f . These are the general objects from calculus that we will work with. We will use time in future lecture's to recall and work with specifics of these definitions e.g., computing the Jacobian of particular functions.

0.5 Probability & Statistics

The overlap between machine learning and statistics is vast. While this course expects at least rudimentary knowledge of statistics and probability theory, this is a field that we will dedicate standard class time towards developing. Like previous sections students should read the following section and find it to be review material.

0.5.1 Random Variables

Typically in statistics, we analyze collected data called samples by formulating a mathematical model of the data-generating process and then inferring quantities about these models from our samples. A critical component of these models are *random variables* which enable us to model uncertain or random events such as a dice roll.

Coin flips or dice rolls are random events whose potential outputs are comprised of a finite collection of events (e.g., heads or tails). For such random events we use a *discrete* random variable whose behavior is described by a probability *mass* function or pmf. For example:

$$f(x; \theta) = \begin{cases} \theta & \text{if } x = 1 \\ 1 - \theta & \text{if } x = 0 \end{cases} \quad (5)$$

This pmf simply assigns the probability $\theta \in [0, 1]$ to the event $x = 1$ and probability $1 - \theta$ to the event $x = 0$. On the otherhand, we often will express probabilities over continuous event spaces. To do this, we use probability density functions or pdfs. The most common example being the normal distribution whose pdf is given by:

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (6)$$

This equation expresses that the probability of an event x is given by the value of the above equation parameterized by μ and σ^2 which we call the mean and variance parameters of the normal distribution.

0.5.2 Moments

While there are an infinite number of probability density and mass functions, we can often describe distributions relatively well with their *moments*. The first moment of a distribution is its mean or center of mass. The second is its variance or the spread of its mass. Though less used in this course, the third movement is the skewness. The first moment or mean of a distribution is also known as the *expectation* of the distribution. We denote the expectation with $\mathbb{E}$ and this can be interpreted as a function of a random variable that returns its average value. Formally this is defined as:

$$\mu = \mathbb{E}[\mathbf{x}] = \int_{-\infty}^{\infty} x p(\mathbf{x} = x) dx \quad (7)$$

We will dive into more specifics on how and why these definitions are what they are when we get to probabilistic machine learning. The standard definition for the n^{th} movement is given as:

$$m_n = \frac{\mathbb{E}[(\mathbf{x} - \mu)^n]}{\mathbb{E}[(\mathbf{x} - \mu)^2]^{n/2}} \quad (8)$$

0.5.3 Independently and Identically Distributed

Independence

In the context of statistical analysis, independence refers to the idea that the values or outcomes of one observation in a dataset do not depend on or influence the values of any other observation. Mathematically, let $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(N)}$ be a set of random variables representing a sample of N observations. These random variables are said to be independent if, for any subset of observations $\mathbf{x}^{(i_1)}, \mathbf{x}^{(i_2)}, \dots, \mathbf{x}^{(i_k)}$ (where $1 \leq i_1 < i_2 < \dots < i_k \leq N$), the joint probability distribution factorizes:

$$P(\mathbf{x}^{(i_1)} = \mathbf{x}^{(i_1)}, \mathbf{x}^{(i_2)} = \mathbf{x}^{(i_2)}, \dots, \mathbf{x}^{(i_k)} = \mathbf{x}^{(i_k)}) = P(\mathbf{x}^{(i_1)} = \mathbf{x}^{(i_1)}) \cdot P(\mathbf{x}^{(i_2)} = \mathbf{x}^{(i_2)}) \cdot \dots \cdot P(\mathbf{x}^{(i_k)} = \mathbf{x}^{(i_k)}) \quad (9)$$

This property is crucial when dealing with real-world data, as it allows us to model each observation as an independent realization of some underlying process. For example, when rolling a fair six-sided die multiple times, the outcomes of each roll are independent events.

Identically Distributed

The "identically distributed" part of i.i.d. refers to the idea that all random variables in the sample follow the same probability distribution. That is, they share the same probability density function (pdf) or probability mass function (pmf), depending on whether the data is continuous or discrete.

Mathematically, if $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(N)}$ are i.i.d. random variables, it means that they all have the same pdf or pmf, denoted as $f_{\mathbf{x}}(\mathbf{x})$, and the cumulative distribution function (cdf) $F_{\mathbf{x}}(\mathbf{x})$.

$$f_{\mathbf{x}^{(1)}}(\mathbf{x}) = f_{\mathbf{x}^{(2)}}(\mathbf{x}) = \dots = f_{\mathbf{x}^{(N)}}(\mathbf{x}) = f_{\mathbf{x}}(\mathbf{x})$$

$$F_{\mathbf{x}^{(1)}}(\mathbf{x}) = F_{\mathbf{x}^{(2)}}(\mathbf{x}) = \dots = F_{\mathbf{x}^{(N)}}(\mathbf{x}) = F_{\mathbf{x}}(\mathbf{x})$$

This assumption simplifies statistical analysis since we can treat each observation in the sample in the same way, knowing they all come from the same distribution.

Bibliography

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